

Quan Xiao | Curriculum Vitae

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EDUCATION

Cornell University (Cornell Tech)

New York, NY

Ph.D. of Electrical and Computer Engineering (GPA: 4.3/4.3)

Aug. 2025 - present

Advisor: Tianyi Chen

Rensselaer Polytechnic Institute

Troy, NY

Ph.D. of Computer and Systems Engineering (GPA: 4.0/4.0)

Sep. 2021 - (Transferred) Aug. 2025

Advisor: Tianyi Chen

University of Science and Technology of China

Hefei, Anhui, China

Bachelor of Statistics

Sep. 2016 - July. 2020

RESEARCH INTEREST

My research focuses on optimization and machine learning, particularly **bilevel and multi-objective optimization**, to develop 1) **efficient methods for multi-goal-oriented generative models**, including reasoning capability, tool use, memory management in LLMs, and the trade-off between training efficiency and exploration; 2) while explicitly accounting for physical constraints of hardware platforms, including **analog in-memory computing (AIMC) for energy-efficient AI training and deployment**. More broadly, I aim to bridge theory and practice by designing scalable and principled methods for safe, reliable, efficient, and hardware-friendly AI systems.

HONORS, AWARDS, AND RESEARCH FUNDING

Google Cloud Research Credits Program (\$1,000 in API credits)

2026 - 2027

NSF ACCESS Award CIS261228: Taming Competing Objectives in Tool-Using LLM Agents

2026 - 2027

Co-Principal Investigator; total award: 750,000 ACCESS GPU credits

Lambda Research Grant (\$2,000 in cloud credits)

2026

ICML Gold Reviewer

2026

Travel Award for NYC Ops Day

2026

NeurIPS Top Reviewer

2025

IEEE Signal Processing Society Scholarship

2024

Belsky Award for Computational Sciences and Engineering in RPI

2023

Neural Information Processing Systems Scholar Award

2023

International Conference on Machine Learning Participation Grant

2022

Successful Participant in Mathematical Contest In Modeling

2019

Excellent student scholarship of the University of Science and Technology of China

2017 - 2019

SELECTED PUBLICATIONS AND PREPRINTS (* equal contribution)

Optimization Theory

- [10] Liuyuan Jiang, **Quan Xiao**, Lisha Chen, and Tianyi Chen. Beyond value functions: Single-loop bilevel optimization under flatness conditions. *Proc. of Neural Information Processing Systems*, 2025.

- [8] Lisha Chen, **Quan Xiao**, Ellen Hidemi Fukuda, Xinyi Chen, Kun Yuan, Tianyi Chen. Efficient Multi-Objective Learning under Preference Guidance: A First-Order Penalty Approach. *Proc. of International Conference on Machine Learning*, 2025. **(Spotlight, top 2.6%)**
- [7] **Quan Xiao**, and Tianyi Chen. Unlocking Global Optimality in Bilevel Optimization: A Pilot Study. *Proc. of International Conference on Learning Representations*, 2025.
- [6] Han Shen, **Quan Xiao**, and Tianyi Chen. On Penalty-based Bilevel Gradient Descent Method. *Mathematical Programming (Series A)*, 2024.
- [5] Liuyuan Jiang, **Quan Xiao**, Victor M Tenorio, Fernando Real-Rojas, Antonio Marques, Tianyi Chen. A Primal-Dual-Assisted Penalty Approach to Bilevel Optimization with Coupled Constraints. *Proc. of Neural Information Processing Systems*, 2024
- [4] **Quan Xiao**, Songtao Lu, and Tianyi Chen. A Generalized Alternating Method for Bilevel Learning under the Polyak-Łojasiewicz Condition. *Proc. of Neural Information Processing Systems*, 2023.
- [3] **Quan Xiao**, Han Shen, Wotao Yin and Tianyi Chen. Alternating Projected SGD for Equality-constrained Bilevel Optimization. *Proc. of International Conference on Artificial Intelligence and Statistics*, 2023.
- [1] Tianyi Chen, Yuejiao Sun, **Quan Xiao**, and Wotao Yin. A Single-Timescale method for stochastic bilevel optimization. *Proc. of International Conference on Artificial Intelligence and Statistics*, 2022. **(Oral, top 2%)**

AI/ML Applications

- [15] Heshan Fernando, **Quan Xiao**, Yan Xin, and Tianyi Chen. ARMOR: Adaptive Retriever Optimization for Low-Resource Telecom Question Answering. arXiv:2606.29706, 2026.
- [14] **Quan Xiao**, Yutong Xuan, Gaowen Liu, Ramana Rao Kompella, and Tianyi Chen. Bilevel Data Curation for LLM Fine-tuning: Offline Selection and Online Self-Refining Generation. arXiv:2511.21056, 2026.
- [9] **Quan Xiao**, Hui Yuan, A F M Saif, Gaowen Liu, Ramana Rao Kompella, Mengdi Wang, Tianyi Chen. A First-order Generative Bilevel Optimization Framework for Diffusion Models. *Proc. of International Conference on Machine Learning*, 2025.
- [2] *Momin Abbas, ***Quan Xiao**, *Lisha Chen, Pin-Yu Chen and Tianyi Chen. Sharp-MAML: Sharpness-Aware Model-Agnostic Meta Learning. *Proc. of International Conference on Machine Learning*, 2022.

Analog In-Memory Computing

- [13] ***Quan Xiao**, *Jindan Li, Zhaoxian Wu, Tayfun Gokmen, and Tianyi Chen. Dynamic Symmetric Point Tracking: Tackling Non-ideal Reference in Analog In-memory Training. *Proc. of International Conference on Machine Learning*, 2026.
- [12] Zhaoxian Wu, **Quan Xiao**, Tayfun Gokmen, Hsinyu Tsai, Kaoutar El Maghraoui, Tianyi Chen. On the Convergence Theory of Pipeline Gradient-based Analog In-memory Training. *IEEE Journal on Selected Areas in Information Theory*, 2026.
- [11] Zhaoxian Wu, **Quan Xiao**, Tayfun Gokmen, Omobayode Fagbohunbe, and Tianyi Chen. Analog In-memory Training on General Non-ideal Resistive Elements: Understanding the Impact of Response Functions. *Proc. of Neural Information Processing Systems*, 2025. **(Oral, top 0.3%)**

INVITED TALK

- **Unlocking Nested Learning for AI: A Case Study on Data Selection for LLM.**
 NYC Operations (Ops) Day, New York, NY. Apr. 2026
 Information Theory and Applications (ITA) Workshop, San Diego, CA. Feb. 2026
- **Global Convergence in Bilevel Optimization: A Pilot Study.**
 Conference on Parsimony and Learning (CPAL), Stanford, CA. March. 2025
 Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA. Nov. 2024

- **Efficient Methods for Bilevel Optimization with Polyak-Lojasiewicz Lower-level Problems.**
Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA. Nov. 2023
SIAM Conference on Optimization (OP23), Seattle, WA. June. 2023

TEACHING EXPERIENCE

- **Teaching Assistant, ECE 5110: Random Signals in Communications and Signal Processing**
Cornell University Spring 2026
- **Guest Lecturer, ECSE 4964/6964: Distributed Optimization & Machine Learning**
Rensselaer Polytechnic Institute Spring 2025
- **Guest Lecturer, ECSE 2500: Engineering Probability**
Rensselaer Polytechnic Institute Fall 2024

INTERNSHIP

- **IBM Thomas J. Watson Research Center, Yorktown, NY.**
Research intern with mentor: Dr. Debarun Bhattacharjya. May. 2024 - Aug. 2024
Project: Uncertainty quantification and selective generation in large language model.
- **IBM Thomas J. Watson Research Center, Yorktown, NY.**
Research intern with mentor: Dr. Songtao Lu. May. 2022 - Aug. 2022
Project: Efficient method for bilevel optimization beyond strong convexity.

VISITING EXPERIENCE

- **Sun Yat-Sen University, Guangzhou, China.**
Host: Prof. Qing Ling. Aug. 2020 – July 2021
- **Michigan State University, East Lansing, MI.**
Host: Prof. Yuying Xie. July 2019 – Oct. 2019

PROFESSIONAL SERVICE

- Session Co-Chair, Asilomar Conference on Signals, Systems, and Computers, 2024.
- Reviewer of Proc. of AAAI ('25), AISTATS ('22 – '26), NeurIPS ('23 – '26), ICLR ('24 – '26), ICML ('24 – '26).
- Reviewer for IEEE Transactions on Signal Processing and IEEE/ACM Transactions on Networking.
- Reviewer for the Journal of Machine Learning Research (JMLR).
- Technical Program Committee Member, Cross-Community Federated Learning Workshop at MLSys, 2022.